

Fingerprinting via the Human Proteome as a Localization Scheme for Nanobots

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Abstract

The localization of nanobots in the human body is a crucial element to enable diagnostic ability. So far, localization schemes mainly rely on mathematical principles to delineate positions of nanobots in the human body. A different approach to determining the position is local pattern recognition, so-called fingerprinting. In this paper, we analyze and evaluate different substances in the environment of the nanobots to find fingerprints for areas of the human body. The human proteome, resembling the entirety of human proteins, grants an inventory with many different genes. A plethora of genes marks different proteins with certain tissue specificities. We present fingerprints using tissue-specific genes for the most significant organs and tissues. With unique gene combinations for each tissue, the human proteome poses a promising localization method for nanobots in the human body.

1 Introduction

To explore the cause of physical discomfort, people often seek the assistance of a physician, consult (online) information sources, or talk to friends and family. While successful treatment at a later stage is feasible for some diseases, certain disorders need to be detected as soon as possible. This early diagnosis might be challenging as the patient could not have any symptoms until later. Persons with chronic illnesses are in even greater need of constant monitoring due to possible immunodeficiency. Regular examinations at the physician can reduce the risk, but even daily visits do not ensure continuous surveillance. Not to mention that daily medical appointments are difficult to reconcile with everyday life. The vision of nanonetworks is that nanodevices (nanoscale devices), for example, patrol the body, take measurements wherever necessary, and send collected data to the outside. Therefore, e.g., a tumor can be detected before it starts metastasizing. Even better, these machines may immediately work on problems they detect within the body, such as cancer cells, arteriosclerosis, or HI viruses. To correctly detect abnormalities, it is essential to connect the measurements of the nanobots to the corresponding organs or tissues in the human body. That requires a reliable localization of the nanobots in question.

Current localization schemes make use of mathematical algorithms like Function Centric Nano-Networking (FCNN) [1]. Using biological or chemical values is an entirely new approach outlining a promising possibility to differentiate particular body regions. Fingerprinting aims to determine the position of nanobots via local pattern recognition. An

individual fingerprint is assigned to each body region by the properties of the environment in question. By identifying these fingerprints, nanobots can determine their position in the human body and communicate it to the outside.

2 Material and Methods

2.1 Approaches

To the best of our knowledge, no method of assigning histological fingerprints to different body regions exists. Therefore, this research study aims to analyze various histological values in the human body regarding their suitability for fingerprinting and find a reliable method.

Possible candidates are systems comprising the whole body, like the lymphatic, the endocrine, and the cardiovascular system.

The lymphatic system must be excluded because nanobots are envisioned to run in the blood circuit. Hence, the values used for fingerprinting have to be present in the cardiovascular system.

The endocrine system and the hormonal balance in the human body include several different parameters which could serve as fingerprinting characteristics. Certain hormones are produced in different glands of the body and then distributed throughout the endocrine system. However, this distribution poses a problem for the clear differentiation of body regions, as there are no single hormones only present in certain regions.

Three methods show favorable characteristics for fingerprinting and are worthy of further analysis, including blood

gases, trace elements, and the human proteome.

2.2 Requirements

To be a reliable source for the localization of nanobots, the parameters used for fingerprinting have to fulfill specific requirements.

For a distinct differentiation between body regions, one possibility is that singular compounds only exist in certain regions. Another possibility would be one compound that shows significantly differentiable concentrations in different regions. For the first option, the existence of the compounds in question should be stable and not influenced by other factors, e.g., diseases or physical activity. The same holds for the concentration of one compound for the second option. It should not be influenced by diseases or physical activity for reliable and continuous monitoring of patients. Another requirement regarding the research of the compound in question is the number of body regions featuring reliable sources. To make reasonable differentiations inside the human body, research for a sufficient number of organs or tissues has to be available. Explicitly critical organs like the heart, lungs, liver, stomach, and kidneys should be covered.

2.3 Suitability

The first compounds under consideration are the gases in the cardiovascular system. A general blood gas analysis measures O_2 , CO_2 , and pH levels. Arteries, veins, and smaller vessels can be distinguished with those characteristics. Oxygen enters the body by breathing and subsequently arrives in the cardiovascular system through the alveolar walls of the lung. It travels to the pulmonary capillaries, binds hemoglobin, and finally reaches tissues, organs, and cells through diffusion. This path occurs inside the arteries, transporting oxygenated blood towards tissues and organs. The veins then transport deoxygenated blood back to the heart. Exceptions are the arteries and veins of the lung, in which case the arteries transport deoxygenated blood and the veins oxygenated blood. Another localization differentiation can be made inside the arteries and veins if one follows the decrease in oxygen concentration towards the capillaries and organs. For a general localization of regions in the human body, this method is not suitable as, e.g., the difference in O_2 concentration could provide information about the distance from the aorta but no clear differentiation of body regions or organs. In addition, physical activity influences O_2 levels in the blood [2] and can therefore not ensure continuous monitoring. The fluctuations in concentration could lead to uncertainties or even wrong diagnostics. Hence, blood gases are excluded as a fingerprinting compound, as the general differentiation of body regions is the primary goal.

Trace elements that occur in minute concentrations in many environments pose another possible fingerprinting characteristic. Essential trace elements can evoke a deficiency syndrome if absent or too low in concentration. The num-

ber of essential elements varies slightly throughout literature, as some are presumably essential but not proven yet. Essential trace elements coinciding throughout literature include Cobalt, Copper, Iodine, Iron, Manganese, Molybdenum, Selenium, and Zinc [3, 4, 5]. Due to the same trace elements being present in different tissues, the concentration of the elements is the focus of interest for fingerprinting. Four of the eight previously mentioned essential trace elements occur in all examined tissues: Copper, Manganese, Selenium, and Zinc. The concentration gradients in different tissues of those four elements can provide insight into the position in the human body. The presence in the respective tissues is the relevant factor for the remaining elements. Reported concentrations for trace elements in different human tissues vary widely [5]. Apart from the variability, the concentrations are minuscule, constituting a problem for the use in fingerprinting due to inadequate reliability. The compounds blood, packed blood cells, and muscle tissue, as reported in [5], cannot be used to identify different body regions. That only leaves urinary excretion, lung tissue, liver tissue, and kidney tissue. These tissues are too few to provide a valuable differentiation between different human body parts. Furthermore, it is impossible to differentiate the concentrations for different ages and sexes, again posing a problem for fingerprinting. Due to those reasons, we omit trace elements as a compound for fingerprinting.

The human proteome represents the entirety of proteins in the human body. It is based on the human genome, the genetic material. Genetic material contains all the information needed to define how an organism is built and is crucial to ensure the functionality and maintenance of a living organism in the way it is intended. The genome describes the entirety of biological information for one organism. To obtain the human proteome from the human genome, individual protein-coding genes (specific sequences of bases) are copied into RNA molecules, called the transcriptome [6]. In the next step, the translation, a polypeptide chain made of amino acids is assembled. Each amino acid is coded by three bases from the RNA molecule. The polypeptide chain is then folded in several steps, acquiring a three-dimensional shape - a protein. The assembly of proteins (gene expression) differs for human tissues and cells, leading to a specific proteome in each cell line and tissue [7]. By measuring the occurrence of particular genes in different tissues, the proteins can be determined. A plethora of research follows up on classifying all human genes according to specificity and occurrence throughout the human body. This information makes it feasible to extract individual fingerprints for different regions via the human proteome.

3 Results and Discussion

The distribution and tissue specificity are taken into account for the choice of protein-coding genes for the fingerprints. Uhlén et al. published a database covering more than 90% of the putative protein-coding genes available at <https://www.proteinatlas.org>. Performing immunohistochemistry on 44 human tissues complemented by RNA-

sequencing on 37 of them, the protein and mRNA expression data was derived [8].

From the approximately 20,000 human protein-coding genes, around 9,000 genes show low tissue specificity and represent the housekeeping proteome [9]. The remaining genes show an elevated expression in particular tissues, subdivided into (i) tissue enriched genes with mRNA levels in one tissue at least four-fold higher than the maximum of any other tissue, (ii) group enriched genes with four-fold higher mRNA levels in a small group of tissues, and (iii) tissue enhanced genes with mRNA levels in a particular tissue at least four-fold higher than the average in all other tissues [9]. As for the distribution, genes can be (i) detected in a single tissue, (ii) detected in some tissues (more than one but less than one-third of all), (iii) detected in many tissues (at least one-third but not all), or (iv) detected in all tissues [9]. The distribution visualizes the number of genes showing detectable mRNA levels in different tissues. Gene detection is done by the normalization of transcriptomics data at hand from several different research groups [9]. A gene has detectable levels of mRNA molecules if the normalized expression (NX value) is above one [9].

For the sample space from which the genes for the fingerprints are chosen, genes not occurring in the respective tissues and genes with low tissue specificity (the housekeeping proteome) are excluded. Table 1 gives an overview of the occurrence of genes according to The Human Protein Atlas in the major organs.

Table 1: Occurrence of Genes in Major Organs

Tissue	Detected	Not Detected	Elevated Genes
Brain	16,227	3,443	2,587
Heart	14,409	5,261	387
Intestine	15,609	4,061	764
Kidney	14,823	4,847	413
Liver	14,110	5,560	936
Lung	15,021	4,649	239
Pancreas	14,490	5,180	422
Stomach	14,707	4,963	159

Furthermore, the elevated genes can be subdivided into four groups regarding the distribution, as mentioned earlier. Genes only detected in a single tissue are advantageous to ensure as unique fingerprints as possible. Since not all organs feature more than one gene from this category, genes detected in some tissues are also considered.

A comparison of the gene data of all relevant tissues showed that none of the tissues have a complete overlap of elevated genes. For our studies, we chose an amount of five genes for each fingerprint. For a reliable localization, the number of genes should be as high as necessary but as low as possible to be still dependable. Upon further research on the prevalence of genes in the examined tissues, the number of genes might differ from five.

The key element for choosing genes for the fingerprints is the normalized expression (NX). The gene expression is estimated by measuring the average transcripts per million (TPM) for different datasets of the human tissues in ques-

tion [9]. For comparability of the different datasets, the numbers are normalized in several steps. The higher the NX value, the higher the mRNA level of a gene in the respective tissue. A *tissue specificity score* compares the tissue with the highest mRNA level of a gene to the tissue with the second-highest mRNA level. Therefore, the remaining genes suitable for the fingerprints are sorted for their *tissue specificity score*. The genes with the highest scores are subsequently chosen for the fingerprints. Table 2 shows the gene combination for the fingerprint of the heart with the corresponding distributions and tissue specificity score.

Table 2: Fingerprint of the Heart

Gene	Tissue Distribution	Tissue Specificity Score
MYL7	Detected in single	822
NPPA	Detected in some	418
TNNI3	Detected in some	367
NPPB	Detected in some	296
TNNT2	Detected in some	241

To guarantee unique fingerprints, we compared the genes chosen for the fingerprints. There is a maximum overlap of one gene between two fingerprints, ensuring unique gene combinations.

4 Conclusion and Outlook

The human body provides a plethora of possibilities to find unique combinations of substances for different body regions. To detect fingerprints via nanobots, the substance in question must be allocated in the cardiovascular system, show significant contrasts in occurrence or concentration in different body regions, and cannot be affected by diseases or physical activity. The human proteome, which resembles the entirety of human proteins, grants an inventory with many different genes which code different proteins with certain tissue specificities. By using the gene specificity and distribution throughout different tissues, unique combinations can be found, accounting for only one body region at a time. Each fingerprint identified in this work consists of two protein-coding genes, showing no intersections in the most crucial genes, ensuring an exclusive localization for each tissue.

The localization of nanobots utilizing histological values is a novel method for which no research is available. In the scope of this work, possibilities, difficulties, and a promising fingerprinting method were pointed out. Many aspects of this method remain open and have to be investigated further in the future.

It is still unclear how nanobots will detect the fingerprints. There are different designs of nanobots which could be electrical or biological. A way of recognizing genes in the human body could be through DNA-tile-based nanobots [10]. Those nanobots consist of assembled DNA molecules while an arbitrary number of DNA strands have sticky ends, meaning that the end of the DNA strand is open and can bind another DNA molecule. With the different designs of

open strands, all logic operators can be applied using DNA-tile-based nanobots. For the 2-gene fingerprints, detection could be realized by a 2-bit-AND tile-based nanobot. Two open strands form counterparts for the fingerprint genes and send a message of localization to the outer environment only if both of the genes bind to the nanobot.

Furthermore, it is worth investigating if only one gene could be sufficient for localization in the human body. The total mRNA concentration and the relative concentration compared to other tissues are provided by the Human Protein Atlas. With this information, concentration gradients can be examined, and conclusions for the localization can be drawn. If the communication of nanobots among each other allows it, the concentrations of different genes can be compared, therefore determining the position of each nanobot.

Depending on the type of nanobot and detection method, the fingerprint genes require more research regarding the biological structure and binding affinity. It has to be ensured that the nanobot binds to the correct genes or measures the correct concentrations, which highly depends on the chemical pull between the chosen nanobot and the gene in question.

A simulation framework for the detection via fingerprints has to be developed to test different scenarios. It would be advantageous to integrate the localization into existing simulations for nanobots in the human body, e.g., BloodVoyagerS which is a body simulator module for the network simulator ns-3 [11]. BloodVoyagerS simulates the blood flow in the human body through the major vessels and integrates nanobots into the flow. It then depicts the distribution of nanobots throughout the human body.

In conclusion, fingerprinting via the proteome is a suitable method to determine positions in the human body. The proteins appearing in different areas can give a definite localization and constitute a promising new method. Further research on the human proteome and the possible detection methods by nanobots has to be carried out to integrate the localization via fingerprinting into existing simulations.

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Conflict of interest: Authors state no conflict of interest.

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